

CHAPTER 1

Introduction to Computer Networks

What a network is · evolution · data communication basics · transmission modes · network types · topologies · protocols & standards

QUICK TRICK

Learn these four things first (details below):

1. **Mesh links** = $n(n-1)/2$; ring and star have **n** links; tree has **n-1** links.
2. **Transfer time** = **file size in BITS** ÷ **data rate in bits per second** (1 byte = 8 bits).
3. **Simplex** = one way · **Half duplex** = both ways, one at a time · **Full duplex** = both ways together.
4. **PAN** < **LAN** < **CAN** < **MAN** < **WAN** in area. Bigger area means lower speed and higher cost.

Most Chapter 1 questions in the exam come from these four ideas plus topologies and basic definitions.

REMEMBER

How these notes match your exam. The BPSC TRE 4.0 Computer Science (class 11-12 / PGT) technical paper follows the school computer science syllabus, where Computer Networks starts with: introduction, evolution of networking, network types, data communication terms, topologies. I built the chapter plan from that syllabus pattern. Please cross-check it with the official BPSC notification before the exam, because I could not open the official PDF itself.

1.1 What is a computer network?

● LEVEL 1 · BEGINNER

The problem. One computer alone can only use what is inside it: its own files, its own printer, its own storage. **The idea.** Join many computers (and printers, phones, servers) with a communication path so they can **share data and resources** and talk to each other.

Definition. A **computer network** is a collection of two or more computers and other devices, connected by a communication medium (wire, fibre or wireless), that exchange data and share resources using agreed rules called protocols.

Analogy. A road network. Cities are the computers, roads are the links, and traffic rules are the protocols. Without roads nothing moves; without rules the traffic crashes.

Term	Meaning	Example
Node	Any device connected to the network that can send, receive or forward data	PC, laptop, printer, router, smartphone
Host (end system)	A node that runs user applications and is the source or destination of data	PC, server, phone
Link	The physical or wireless path that joins two nodes	UTP cable, optical fibre, Wi-Fi radio link
Server	A powerful computer that gives a service or resource to others	Web server, file server, mail server
Client	A computer that requests the service	Your browser on a laptop
Network interface card (NIC)	Hardware that connects a device to the network; has a unique 48-bit MAC address	Ethernet card, Wi-Fi card

QUICK TRICK

Every host is a node, but not every node is a host. A router or switch is a node, but it only forwards data; it is not the final user of the data. **Minimum devices needed for a network = 2.**

1.2 Why do we need networks? Goals, advantages and disadvantages

● LEVEL 1 • BEGINNER

Goal / Advantage	Explanation
Resource sharing	Hardware (printer, scanner, disk), software (one licensed package) and data are shared, so each user does not buy their own.
Communication	Email, chat, voice and video calls, file transfer. Fast and cheap compared with post.
Reliability	Important data can be copied on more than one machine. If one machine fails another can take over.
Cost saving	Sharing expensive devices and software lowers the total cost.
Scalability	New computers can be added without redesigning everything.
Central management and backup	One administrator can control users, security and backup from one place.
Remote access	Work or study from another city using the same data.
Higher processing power	Many computers can work together on one big job (distributed / cluster computing).
Disadvantage	Explanation
Security risk	Hackers can attack from anywhere; one infected machine can spread malware to many.
Setup and maintenance cost	Cables, devices and a skilled administrator are needed.
Dependence on central devices	If the server, switch or main cable fails, many users are stopped.
Privacy problems	Data moving on shared links can be seen or stolen if not protected.

Where are networks used?

- **Business:** online shopping, ATM and UPI payments, internet banking, video meetings.
- **Education:** e-learning platforms, online classes, digital libraries (for example SWAYAM and DIKSHA in India).
- **Government:** e-governance services, online forms, DigiLocker, railway reservation.
- **Health:** telemedicine, online reports, hospital information systems.
- **Home and personal:** streaming, social media, smart-home devices (Internet of Things).

1.3 Evolution of networking

● LEVEL 1 • BEGINNER

Networks grew from a few connected research computers to a network of billions of devices. Know this timeline well; year-based questions are common.

Year	Event	Remember
1960s	Packet switching theory (Paul Baran, Donald Davies, Leonard Kleinrock)	Data is cut into small packets
1969	ARPANET started by ARPA (Advanced Research Projects Agency) of the US Department of Defense. First 4 nodes: UCLA, Stanford Research Institute (SRI), UC Santa Barbara, University of Utah. First message was sent from UCLA to SRI on 29 October 1969.	First packet-switched network; ancestor of the Internet
1971	First network email by Ray Tomlinson; he chose the @ sign	@ sign for email addresses
1973	Ethernet idea by Robert Metcalfe at Xerox PARC (later standardised as IEEE 802.3)	LAN technology
1974	TCP concept published by Vint Cerf and Bob Kahn	Cerf and Kahn = fathers of the Internet
1 Jan 1983	ARPANET officially switched to TCP/IP	Birth of the Internet
1986	NSFNET backbone (US National Science Foundation) joins universities	Backbone for academic Internet
1989–91	World Wide Web by Tim Berners-Lee at CERN: proposal 1989, first website and public release 1991	WWW, HTTP, HTML, URL
1990	ARPANET was shut down; its role was taken over by the Internet	
1993	Mosaic, the first popular graphical web browser (NCSA)	Browser boom

Year	Event	Remember
15 Aug 1995	VSNL started public Internet access in India	Internet in India
2022	5G services launched in India (1 October 2022)	5G

BPSC POINT

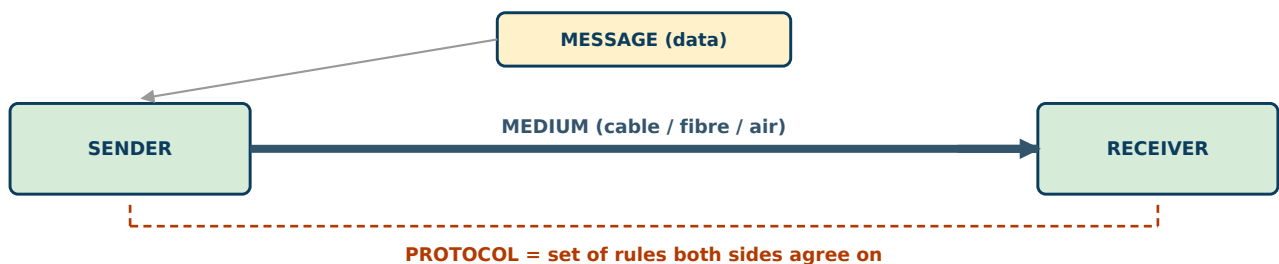
Interspace (as in the school textbook): a client/server software program that lets many users communicate online with real-time audio, video and text chat in dynamic 3D environments, giving the feeling of a virtual world.

ARPANET = 1969 · TCP/IP adopted = 1983 · WWW = 1989-91 · Internet in India (VSNL) = 1995.

1.4 Data communication and its five components

● LEVEL 1 · BEGINNER

Data communication means exchange of data between two devices through a transmission medium. It needs five components. If any one is missing, communication fails.



Component	Meaning	Example
Message	The information to be sent: text, number, image, audio or video	An email, a photo
Sender	The device that sends the message	Your computer, a phone
Receiver	The device that receives the message	Friend's phone, server
Transmission medium	The physical path from sender to receiver	Twisted pair, coaxial cable, optical fibre, radio waves
Protocol	Set of rules that both sender and receiver follow	HTTP, TCP/IP, SMTP

QUICK TRICK

Memory line: **M-S-R-M-P** = Message, Sender, Receiver, Medium, Protocol. Note: the **protocol** is the part people forget; two devices with a cable but no common protocol cannot understand each other.

1.5 Data communication terms: channel, bandwidth, data rate, throughput, latency

● LEVEL 2 · EXAM

These words look similar but have different meanings. Examiners like to mix them.

Term	Meaning	Unit
Channel	The path (physical or logical) through which the signal travels from sender to receiver	—
Bandwidth (analog sense)	Range of frequencies a channel can pass = highest frequency – lowest frequency	Hz, KHz, MHz, GHz
Bandwidth (digital sense)	Maximum amount of data a channel can carry per second	bps, Kbps, Mbps, Gbps, Tbps
Data (transfer) rate	Number of bits actually sent per second	bps
Throughput	Real data rate achieved in practice. Always $\leq$ bandwidth because of delays, errors and traffic	bps
Latency (delay)	Time taken for data to travel from source to destination	ms, s
Baud rate	Number of signal changes (signal elements) per second	baud

Analogy. Water pipe. Bandwidth = width of the pipe. Throughput = water that actually comes out per second (less than the pipe can carry if it is partly blocked). Latency = time the first drop takes to reach the end.

Units of data rate

Unit	Value (networking / SI style)	Note
1 Kbps	1 000 bps	
1 Mbps	1 000 Kbps = 10^6 bps	Internet plans are in Mbps
1 Gbps	1 000 Mbps = 10^9 bps	
1 Tbps	1 000 Gbps = 10^{12} bps	
1 byte	8 bits	Small b = bit, capital B = byte

COMMON MISTAKE

Some school textbooks use 1 Kbps = 1024 bps. Real networking and GATE use 1000. If an exam question gives both styles as options, follow the book your syllabus uses; if it gives only one numeric answer, check which style it fits.

Delay formulas (Level 3 idea, but easy)

Delay	Formula	Meaning
Transmission delay	$T_t = L \div R$	L = packet length in bits, R = link rate in bps. Time to push all bits onto the link.
Propagation delay	$T_p = d \div s$	d = distance, s = signal speed (about 2×10^8 m/s in copper or fibre). Time for one bit to travel.
Queuing delay	(depends on traffic)	Time waiting in the router buffer.
Processing delay	(small)	Time to check headers and decide the next hop.
Total (one hop)	$T_t + T_p + \text{queuing} + \text{processing}$	
Bandwidth-delay product	Rate $\times$ propagation delay	Number of bits that can be "in the pipe" at one time.
Bit rate	Baud rate $\times$ bits per signal element	If one signal element carries 3 bits, bit rate = $3 \times$ baud.

Worked examples

Example 1. A 25 MB file is sent over a 10 Mbps link. Find the time (ignore delays and take 1 MB = 10^6 bytes).
Size in bits = $25 \times 8 = 200$ Mb. Time = $200 \text{ Mb} \div 10 \text{ Mbps} = \mathbf{20 \text{ s}}$.

Example 2. A channel passes frequencies from 20 KHz to 100 KHz. Bandwidth = $100 - 20 = \mathbf{80 \text{ KHz}}$.

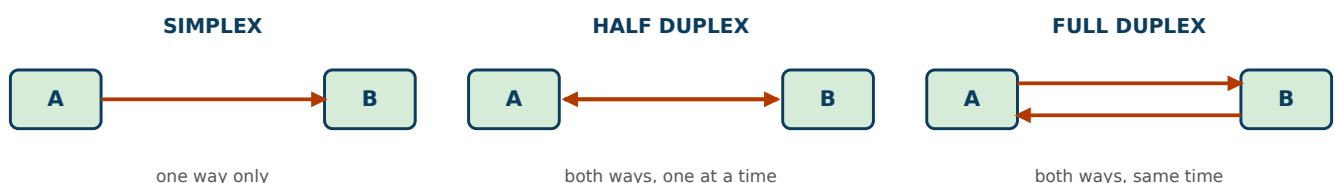
Example 3. A packet of 1500 bytes goes on a 100 Mbps link of length 2000 km (signal speed 2×10^8 m/s).
L = $1500 \times 8 = 12\,000$ bits. $T_t = 12\,000 \div 10^8 = 1.2 \times 10^{-4} \text{ s} = \mathbf{120 \mu\text{s}}$.
 $T_p = 2 \times 10^6 \text{ m} \div 2 \times 10^8 \text{ m/s} = 0.01 \text{ s} = \mathbf{10 \text{ ms}}$. Total $\approx \mathbf{10.12 \text{ ms}}$.

COMMON MISTAKE

Bandwidth in Hz (width of frequency band) and bandwidth in bps (data capacity) are two different uses of the same word. In a question, look at the unit: Hz means frequency range, bps means data capacity.

1.6 Transmission modes (direction of data flow)

● LEVEL 2 · EXAM



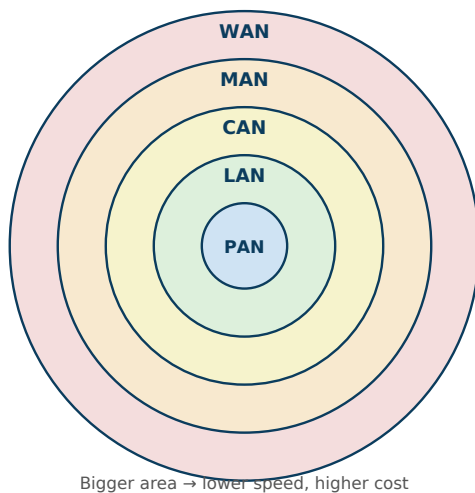
Mode	Direction	Examples	Capacity use
Simplex	One direction only. Receiver can never reply.	Keyboard → computer, TV and radio broadcast, mouse, monitor	Full channel used one way
Half duplex	Both directions, but only one side at a time.	Walkie-talkie, CB radio, old Ethernet with a hub	Channel shared in turns
Full duplex	Both directions at the same time.	Telephone, mobile call, modern switched Ethernet	Can use two-way capacity together

QUICK TRICK

Think of roads. Simplex = one-way street. Half duplex = single-lane bridge where cars take turns. Full duplex = two-lane road with a lane for each direction.

Serial and parallel transmission (how the bits travel)

	Serial	Parallel
How	Bits go one after another on one line	Several bits (for example 8) go together on separate lines
Speed over short distance	Slower	Faster
Cost and wiring	Cheap, one wire	Costly, many wires
Distance	Good for long distance (used in networks, USB)	Only short distance (inside computer, old printer port) because bits arrive at slightly different times

1.7 Types of networks by size (geographical area)● **LEVEL 2 · EXAM****Read the picture from inside to outside.**

PAN → LAN → CAN → MAN → WAN. Each ring covers a larger area than the one inside it.

As the area grows: **speed goes down, error rate goes up, cost goes up**, and ownership changes from one person or organisation (private) to many carriers (public / leased).

Type	Full form	Range (approx.)	Details	Examples
PAN	Personal Area Network	1 – 10 m	Around one person. Wired (USB) or wireless (Bluetooth, NFC, infrared).	Phone ↔ earbuds, laptop ↔ printer by Bluetooth
LAN	Local Area Network	Up to about 1–2 km	One building or office. Owned by one organisation. High speed (100 Mbps to 10 Gbps), low error rate. Ethernet and Wi-Fi.	School computer lab, office network
CAN	Campus Area Network	1 – 5 km	Several linked LANs inside a campus of one organisation.	University or college campus
MAN	Metropolitan Area Network	About 5 – 50 km (a city)	Joins LANs across a city. Often owned by a service provider or city authority.	Cable TV network, city-wide Wi-Fi, bank branches in one city
WAN	Wide Area Network	Country, continent, world	Uses leased lines, optical backbone, satellite. Slower and costlier than LAN. Often a network of LANs and MANs.	The Internet (largest WAN), bank ATM network, railway network

Point	LAN	MAN	WAN
Area	Small (building)	City	Country / world
Speed	Highest	Medium	Lowest
Error rate	Lowest	Medium	Highest
Ownership	Private (one organisation)	Private or public	Mostly public / leased
Setup cost	Lowest	Medium	Highest
Typical media	UTP, Wi-Fi	Fibre, microwave	Fibre, satellite, leased line

QUICK TRICK

Quick match: **Bluetooth** = **PAN** · **School lab** = **LAN** · **Cable TV** = **MAN** · **Internet** = **WAN**.

Other names you may see: **WLAN** (wireless LAN, IEEE 802.11), **HAN** (home area network), **VPN** (private tunnel over the public Internet), **SAN** (storage area network).

1.8 Other ways to classify networks

● LEVEL 2 · EXAM

(a) By architecture: client-server and peer-to-peer

Point	Client-Server	Peer-to-Peer (P2P)
Roles	Fixed: servers give service, clients ask	Every computer is both client and server
Control	Centralised administration, central user accounts	No central control; each computer is managed by its owner
Security	Better and easier to enforce	Weaker
Cost	Higher (server hardware, administrator)	Low
Failure	Server failure affects everybody	One peer failing does not stop the others
Best size	Medium to large networks	Small (about 10 computers or fewer)
Examples	Web, email, school server	File sharing between two home PCs, BitTorrent

(b) By transmission technology

Type	How it works	Example
Broadcast network	One shared communication channel; a message sent by one machine is received by all, and each checks if it is for itself	Wi-Fi, old hub-based Ethernet
Point-to-point network	Many separate links, each joining exactly one pair of machines. Data may pass through several nodes to reach the destination.	Switched networks, WAN leased lines

(c) By how many receivers get the message

Mode	Sender → Receivers	Example
Unicast	One → one specific receiver	Opening a web page
Multicast	One → a chosen group	Live video lecture for a group of subscribers
Broadcast	One → all devices in the network	ARP request, TV broadcast

(d) By medium and ownership

- **Wired network:** uses cables (twisted pair, coaxial, fibre). Stable and fast.
- **Wireless network:** uses radio waves, microwaves, infrared. Flexible but easier to intercept and affected by interference.
- **Private network** belongs to one organisation; **public network** is open to everyone (for example the Internet).

1.9 Internet, Intranet, Extranet and the World Wide Web

● LEVEL 1 · BEGINNER

	Internet	Intranet	Extranet
Meaning	Global network of networks that uses TCP/IP	Private network of one organisation that uses Internet technology	Intranet that is shared in a limited way with outside partners
Access	Open to anyone	Only members of the organisation	Employees plus approved outsiders (suppliers, customers)
Example	Browsing a news website	College staff portal, company mail system	Supplier order-tracking portal

	Internet	World Wide Web (WWW)
What is it	The huge physical and logical network of connected computers (the infrastructure)	A service on the Internet: a system of linked web pages, accessed by a browser using HTTP
Invented	ARPANET era (1969), TCP/IP in 1983	Tim Berners-Lee, 1989-91
Other services on the Internet	Email, FTP, VoIP, online games	—

ISP (Internet Service Provider) is a company that gives you access to the Internet (for example BSNL, Jio, Airtel). ISPs are arranged in levels: Tier-1 ISPs own global backbones, Tier-2 ISPs buy from Tier-1 and sell to Tier-3, and Tier-3 (local) ISPs connect homes and offices.

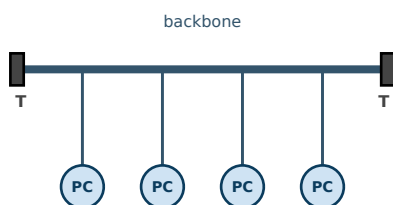
COMMON MISTAKE

The Internet and the Web are not the same. **Internet = the roads. Web = one kind of traffic on those roads.** Email still works without the Web.

1.10 Network topologies

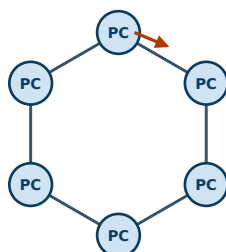
● LEVEL 2 · EXAM

Topology is the way devices (nodes) and links are arranged in a network. **Physical topology** is the real layout of cables and devices. **Logical topology** is how data actually flows. The two can differ: an old Ethernet built with a hub is a **physical star** but works like a **logical bus**, because the hub repeats every signal to every port.



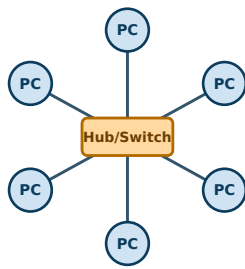
1. Bus topology

- One long **backbone cable**; each node joins through a **drop line** / **T-connector**. Both ends have a **terminator** that absorbs the signal so it does not bounce back.
- Cable needed: 1 backbone + n drop lines. Access method: CSMA/CD (coaxial Ethernet 10Base2, 10Base5).
- **Pros**: cheap, least cable, easy for a very small network.
- **Cons**: a break in the backbone stops the whole network; collisions rise with traffic; hard to find faults; limited length and nodes.



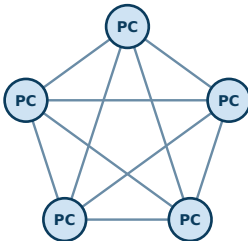
2. Ring topology

- Each node is joined to exactly **two neighbours**, forming a closed loop. Data moves in one direction (usually) and each node passes it on. **Links = n**.
- Access by **token passing** (IEEE 802.5 Token Ring). FDDI uses a **dual ring** so it survives one break.
- **Pros**: orderly access, no collisions, every node gets a fair turn.
- **Cons**: in a single ring one node or cable failure can stop the whole ring; adding or removing a node disturbs the network; data passes through many nodes so it can be slow.



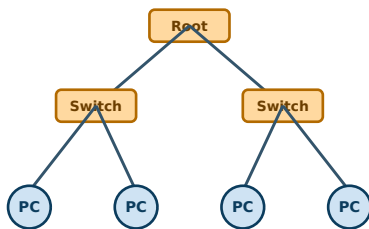
3. Star topology

- Every node connects to a **central hub or switch**; nodes never connect directly to each other. **Links = n**.
- Today's Ethernet LANs are physically star (UTP cable to a switch).
- **Pros**: easy to install and extend; failure of one cable or node affects only that node; easy fault finding.
- **Cons**: the central device is a **single point of failure**; needs more cable than bus.



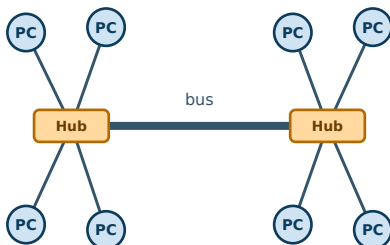
4. Mesh topology

- **Full mesh**: every device has a dedicated link to every other device. **Links = $n(n-1)/2$** and **ports per device = $n-1$** .
- **Partial mesh**: only some devices are fully connected. Wireless mesh and WAN backbones use this idea.
- **Pros**: no traffic jam on dedicated links; very reliable (another path exists); private and secure; easy to isolate a fault.
- **Cons**: very costly (cables and ports); installation and re-configuration are difficult.



5. Tree (hierarchical) topology

- A **root** device at the top, branches below, like an upside-down tree. It is a combination of star topologies joined in a hierarchy (star-bus).
- For n devices (counting hubs and switches too) **links = $n-1$** .
- **Pros**: easy to expand; each branch can be managed separately; fault in one branch is isolated.
- **Cons**: if the root or an upper device fails, everything below loses connection; more cable and more complex.



6. Hybrid topology

- A mix of **two or more** different topologies, for example star-bus or star-ring. Large real networks (a campus or a company) are hybrid.
- **Pros**: flexible, can be designed to suit each department, easy to grow.
- **Cons**: design and fault-finding are complex; costly.

Number of links and ports for n devices

Topology	Links (cables)	n = 4	n = 5	n = 8	n = 10
Bus	1 backbone + n drop lines	5	6	9	11
Ring	n	4	5	8	10
Star	n	4	5	8	10
Tree	$n - 1$ (all nodes counted)	3	4	7	9
Full mesh	$n(n-1)/2$	6	10	28	45

BPSC POINT

Full mesh with duplex links needs $n(n-1)/2$ cables. If each link is simplex (one-way) you need $n(n-1)$.

Each device in a full mesh needs $(n-1)$ I/O ports. Adding the $(n+1)^{\text{th}}$ device to a full mesh adds n new links.

Comparison of topologies

Topology	If something fails	Cost	Cable	Access / traffic	Where used
Bus	Backbone break stops all	Lowest	Least	Shared; collisions	Old coax LANs, small labs
Ring	One break can stop all (single ring)	Medium	Medium	Token, no collision	Token Ring, FDDI, SONET rings
Star	Only that node; hub failure stops all	Medium	More than bus	Via central device	Almost every modern LAN
Mesh	Traffic takes another path	Highest	Most	Dedicated links	WAN / backbone, critical links
Tree	Branch below failed device is cut off	Med-High	More than star	Hierarchical	Large campus, organisation

QUICK TRICK

Which topology? Think failure first.

Whole network dies when ONE cable breaks → **bus**. Whole network dies when the CENTRAL device breaks → **star**.

Network keeps working even if one link breaks → **mesh**. Token used → **ring**. Hierarchy with a root → **tree**.

COMMON MISTAKE

- Using $n(n-1)$ for a normal full mesh. Duplex mesh is $n(n-1)/2$.
- Saying a hub-based Ethernet is a logical star. It is a physical star but a logical bus.
- Forgetting the terminators in bus topology. Without them the signal reflects and corrupts data.

1.11 Protocols and standards

● LEVEL 1 • BEGINNER

Protocol: a set of rules that decides **how** two devices communicate. Every protocol has three key elements:

Element	Meaning	Example
Syntax	Structure or format of the data (what order the fields come in)	First 8 bits = sender address, next 8 = receiver address
Semantics	Meaning of each field: what action to take	An address field tells which route to use
Timing	When to send and how fast	A fast sender must slow down for a slow receiver

Standards are agreed rules that let products of different companies work together. **De facto** standards become standard by common use (not officially approved); **de jure** standards are approved by an official body.

Organisation	Role
ISO (International Organization for Standardization)	Created the OSI reference model
ITU-T (earlier CCITT)	Telecom and data communication standards (for example V.90 modems, X.25)
IEEE	LAN standards, the 802 series
IETF (Internet Engineering Task Force)	Internet protocols; publishes RFC (Request for Comments) documents
ANSI, EIA/TIA	National standards; EIA/TIA for cabling
ICANN / IANA	Domain names, IP address and port number allocation
W3C	Web standards such as HTML and CSS

IEEE 802 standards to remember

Standard	What it covers
802.3	Ethernet (wired LAN, CSMA/CD)
802.11	Wi-Fi (wireless LAN)
802.15	Bluetooth and other wireless PAN
802.16	WiMAX (wireless MAN)
802.5	Token Ring
802.4	Token Bus
802.1	LAN architecture and bridging

QUICK TRICK

Memory: **3 = wired Ethernet, 11 = Wi-Fi, 15 = Bluetooth, 16 = WiMAX, 5 = Token Ring.**

1.12 Common mistakes

● LEVEL 2 · EXAM

COMMON MISTAKE

- Mixing **Mbps** (megabits per second) and **MBps** (megabytes per second). 1 MBps = 8 Mbps.
- Treating bandwidth and throughput as the same. Throughput is the real speed and is always lower or equal.
- Saying Internet and WWW are the same thing.
- Confusing half duplex with simplex. Half duplex allows both directions, just not together.
- Using the wrong link formula for a mesh, or forgetting that the tree formula is $n-1$ only when all devices are counted.
- Calling a bus topology "fast". It is cheap, not fast; collisions slow it down.
- Forgetting that LAN, MAN, WAN are ordered by area, not by importance.

Quick-Solve Tricks — Summary

Situation	Trick
Network definition	2 or more devices + medium + protocol = network
Five components	Message, Sender, Receiver, Medium, Protocol
File transfer time	Size in bits ÷ rate in bps (multiply bytes by 8)
Analog bandwidth	$f_{\text{high}} - f_{\text{low}}$ (in Hz)
Bit rate	Baud × bits per signal element
Delays	$T_t = L/R$, $T_p = d/s$
Direction of flow	Simplex (one way), half duplex (turns), full duplex (together)
Area order	PAN < LAN < CAN < MAN < WAN
Full mesh	Links $n(n-1)/2$, ports per device $n-1$
Ring or star	n links
Tree	$n-1$ links
Bus	1 backbone + n drop lines; terminators at both ends
Hub-based Ethernet	Physical star, logical bus
Dates	ARPANET 1969 · TCP/IP 1983 · WWW 1989-91 · VSNL India 1995

BPSC QUICK CHECK — Chapter 1

ARPANET → 1969	TCP/IP adopted → 1 Jan 1983
Internet = network of networks	WWW = a service on the Internet (Tim Berners-Lee)
Bluetooth → PAN	Cable TV → MAN, Internet → WAN
Telephone → full duplex	Walkie-talkie → half duplex, TV broadcast → simplex
Mesh links = $n(n-1)/2$	Star and ring links = n , tree links = $n-1$
802.3 Ethernet · 802.11 Wi-Fi	802.15 Bluetooth · 802.16 WiMAX

Traps: bits vs bytes · bandwidth vs throughput · Internet vs Web · physical vs logical topology · half duplex vs simplex · duplex mesh uses $n(n-1)/2$ not $n(n-1)$.

Practice questions

● LEVEL 1 · BEGINNER

- **Q1.** Define a computer network and name any two advantages.
- **Q2.** Which network was the forerunner of the Internet, and in which year did it start?
- **Q3.** Name the five components of data communication.
- **Q4.** Which transmission mode is used in a telephone conversation?
- **Q5.** Expand PAN, LAN, MAN and WAN.
- **Q6.** In which topology is a central hub or switch used?

● LEVEL 2 · EXAM

- **Q7.** How many links are needed to connect 8 computers in a full mesh? How many ports does each computer need?
- **Q8.** A 50 MB file is sent on a 20 Mbps link. How long does it take? (1 MB = 10^6 bytes, ignore delays.)
- **Q9.** A channel passes frequencies from 2 MHz to 5.5 MHz. Find its bandwidth.
- **Q10.** A signal has 2000 baud and each signal element carries 3 bits. Find the bit rate.
- **Q11.** What happens in a bus topology if the backbone cable breaks? Why are terminators needed?
- **Q12.** Give one difference between the Internet and the World Wide Web.

● LEVEL 3 · BPSC / PGT

- **Q13.** A 1000-byte packet is sent over a 1 Mbps link of length 3000 km (signal speed 2×10^8 m/s). Find the transmission delay, propagation delay and their total (ignore queuing and processing).
- **Q14.** A link has bandwidth 100 Mbps and propagation delay 10 ms. Find the bandwidth-delay product.
- **Q15.** A tree network has one root switch, 4 second-level switches, and 5 computers on each second-level switch. Find the total number of devices and the total number of links.
- **Q16.** A fully meshed network of n devices gets one more device. How many new links and how many new ports per existing device are needed?
- **Q17.** Why is an Ethernet LAN built with a hub called a physical star but a logical bus?

Answers

Q	Solution
1	A network is two or more devices connected by a medium to share data and resources using protocols. Advantages: resource sharing, faster communication (also reliability, cost saving).
2	ARPANET, 1969.
3	Message, sender, receiver, transmission medium, protocol.
4	Full duplex (both people can talk and listen at the same time).
5	Personal Area Network, Local Area Network, Metropolitan Area Network, Wide Area Network.
6	Star topology (also the lower levels of a tree).
7	Links = $n(n-1)/2 = 8 \times 7 \div 2 = \mathbf{28}$. Ports per computer = $n - 1 = \mathbf{7}$.
8	50 MB = $50 \times 8 = 400$ Mb. Time = $400 \div 20 = \mathbf{20\ s}$.
9	$5.5 - 2 = \mathbf{3.5\ MHz}$.
10	Bit rate = $2000 \times 3 = \mathbf{6000\ bps}$.
11	The backbone is the only shared path, so the network splits or stops working. Terminators absorb the signal at the ends so that it does not reflect back and cause errors.
12	The Internet is the global network of networks (infrastructure). The WWW is a service on it: web pages linked by hyperlinks and accessed with a browser over HTTP. Email and FTP use the Internet but are not the Web.
13	$L = 1000 \times 8 = 8000$ bits. $T_t = 8000 \div 10^6 = \mathbf{8\ ms}$. $T_p = 3 \times 10^6 \div 2 \times 10^8 = 0.015\ s = \mathbf{15\ ms}$. Total = $\mathbf{23\ ms}$.
14	$BDP = 100 \times 10^6\ bps \times 0.01\ s = 10^6$ bits = $\mathbf{1\ Mb}$ (125 000 bytes).
15	Devices = $1 + 4 + (4 \times 5) = \mathbf{25}$. Links = 4 (root to switches) + 20 (switches to PCs) = $\mathbf{24}$, which equals $n - 1 = 25 - 1$ ✓.
16	With n devices already meshed, the new $(n+1)^{th}$ device must be linked to all n : $\mathbf{n\ new\ links}$. Every existing device needs $\mathbf{1\ extra\ port}$ (from $n-1$ to n).
17	Physically every PC has its own cable to the hub, so it looks like a star. But the hub repeats each signal to all ports, so all stations share one medium and collide like on a bus, so logically it is a bus.

REMEMBER

Next: Chapter 2 — Transmission Media (twisted pair, coaxial, optical fibre, radio waves, microwaves, infrared, satellite, wired vs wireless).

Planned roadmap: Ch 3 Network Devices · Ch 4 Switching, Multiplexing and Error Detection · Ch 5 OSI and TCP/IP Models · Ch 6 Network Protocols (HTTP, FTP, SMTP, POP3, TELNET, PPP, VoIP, DNS) · Ch 7 IP Addressing and Subnetting · Ch 8 Web Services (WWW, HTML, XML, URL, domain, hosting) · Ch 9 Network Security · Ch 10 Mobile Telecom (1G-5G), Cloud and IoT.